

P5 IS, strategija in vrednost

OIS 2025/2026

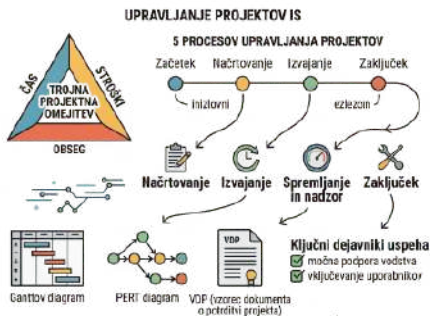
izr. prof. dr. Dejan Lavbič

18. 3. 2026

Vsebina predavanj

- **Rainer (2023)**, Introduction to Information Systems: Supporting and Transforming Business,
 - **Ch. 13: Acquiring Information Systems and Applications**
- **Wallace (2021)**, Introduction to Information Systems,
 - **Ch. 12: Information Systems Project Management and Strategic Planning**
- **Laudon (2023)**, Essentials of MIS,
 - **Ch. 12: Making the Business Case for Information Systems and Managing Projects**

Kaj si bomo danes pogledali?



Strategija in načrtovanje IS

Tehnični dolg in poslovna vrednost IS

- Primer: **Southwest Airlines (2022)**,
- **zastarela IT infrastruktura**,
- **nezmožnost skaliranja operacij**,
- **17.000 odpovedanih letov**,
- posledica: **tehnični dolg**.



Načrtovanje IS



- **Organizacijska strategija → IT strategija,**
- **povezava: poslovni cilji – informacijski sistemi,**
- **rezultat: IS razvojni projekti.**

Namen načrtovanja IS



Kaj je **najpomembnejši namen načrtovanja** informacijskih sistemov?

- **Dolgoročni cilji IT,**
- podpora **poslovni strategiji,**
- usmerjanje **IT arhitekture,**
- razporeditev **razvojnih virov.**

Strateško načrtovanje informacijskih sistemov

Model	Description	% in use
Subsidy model	No chargebacks; costs are allocated to business units	30%
Cost center model	Similar to subsidy, but some chargebacks are in place	30%
Service center model	Chargebacks for IT services	30%
Profit center model	Services billed at market rates to help fund new IT projects	10%

Strateški načrt IS vključuje

- Vizijo in načela uporabe IT,
- upravljanje portfelja projektov,
- arhitekturo informacijskih sistemov,
- obnovo po katastrofah (*angl. disaster recovery*),
- tehnološke in poslovne trende.

Načrt informacijskih sistemov

- Namen,
- strateška utemeljitev,
- trenutni sistemi,
- novi projekti,
- strategija upravljanja,
- implementacija,
- proračun.

Poslovna vrednost IT investicij

- Odločanje: **stroški vs. koristi**,
- metoda: **analiza stroškov in koristi** (*angl. cost-benefit*),
- izziv: **težko oceniti koristi**,
- primeri metod:
 - **neto sedanja vrednost** (*angl. Net Present Value (NPV)*),
 - **donosnost investicije** (*angl. Return on Investment (ROI)*),
 - **analiza točke pokritja** (*angl. Break-even analysis*),
 - **poslovni primer** (*angl. Business Case*).

Kaj je najtežje oceniti?



Kaj je pri opravičevanju investicije IT **običajno najtežje oceniti?**

Stroški in koristi informacijskih sistemov

Costs

- Hardware
- Networking
- Software
- Services
- Personnel

Tangible Benefits (Cost Savings)

- Increased productivity
- Lower operational costs
- Reduced workforce
- Lower computer expenses
- Lower outside vendor costs
- Lower clerical and professional costs
- Reduced rate of growth in expenses
- Reduced facility costs

Intangible Benefits

- Improved asset utilization
- Improved resource control
- Improved organizational planning
- Increased organizational flexibility
- More timely information
- Improved customer experience
- Increased organizational learning
- Legal requirements attained
- Enhanced employee goodwill
- Increased job satisfaction
- Improved decision making
- Higher client satisfaction
- Better corporate image

Finančna analiza IT investicij

	A	B	C	D	E	F	G	H
1	Estimated Costs & Benefits - Mobile Online Ordering System							
2	Year		0	1	2	3	4	5
3			2022	2023	2024	2025	2026	2027
4	Costs							
5	Hardware							
6	50 iPads @\$500		\$25,000					
7	Cloud IaaS		\$4,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000
8	Networking		\$1,500	\$1,500	\$1,500	\$1,500	\$1,500	\$1,500
9	Software							
10	Mobile ordering app		\$35,000					
11	Integration with ERP		\$25,000					
12	Human Resources							
13	Business Staff		\$10,000					
14	IT Staff + Consultants		\$45,000	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000
15	Training		\$7,000	\$1,000	\$1,000	\$1,000	\$1,000	\$1,000
16	Maintenance and Support			\$5,000	\$5,000	\$5,000	\$5,000	\$5,000
17	Annual Costs		\$152,500	\$11,500	\$11,500	\$11,500	\$11,500	\$11,500
18	Total Costs		\$210,000					
19	Benefits							
20	Reduced labor costs			\$52,000	\$52,000	\$52,000	\$52,000	\$52,000
21	Reduced errors and returns			\$70,000	\$70,000	\$70,000	\$70,000	\$70,000
22	Annual Net Cash Flow		-\$152,500	\$110,500	\$110,500	\$110,500	\$110,500	\$110,500
23	Total Benefits		\$400,000					
24								
25	Net Present Value		\$268,407					
26	ROI		4.1%					
27	Internal Rate of Return		17.0%					

Model ocenjevanja (*angl. Scoring model*)

Criteria	Weight	Alternative 1 (%)	Alternative 1 Score	Alternative 2 (%)	Alternative 2 Score
1.1 Order processing					
1.2 Online order entry	5	67	335	83	415
1.3 Order tracking by sales rep	5	81	405	75	375
1.4 Order tracking by customer	5	30	150	80	400
Total order processing			890		1,190
2.1 Ease of use					
2.2 System access from mobile platforms	5	55	275	92	460
2.3 Short training time	4	79	316	85	340
2.4 User-friendly online screens and data entry	4	65	260	87	348
Total ease of use			851		1,148
3.1 Costs					
3.2 Software costs	3	51	153	65	195
3.3 Hardware (cloud services) costs	4	57	228	90	360
3.4 Maintenance and support costs	4	42	168	89	356
Total costs			549		911
Grand Total			2,290		3,249

Pristop z opcijami pri naložbah v IT (1/3)

- Odločitev o uvedbi novega **ERP sistema**.
- Sistem ima **5 modulov**, vsak stane **10 mio €**.
- Uspeh je odvisen od **sprejetja med zaposlenimi**.
- Ob uspehu prinese **200 mio € koristi**.
- Ob neuspehu le **10 mio € koristi**.
- Ocenjena **verjetnost uspeha: 50 %**.

Pristop z opcijami pri naložbah v IT (2/3)

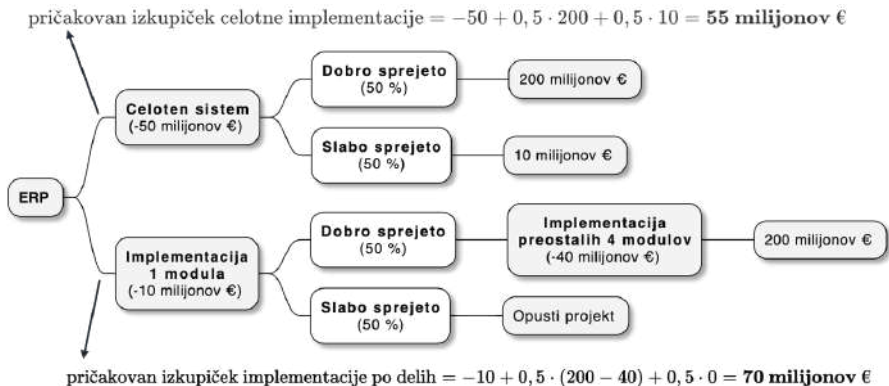
- Vodja informatike ima **2 možnosti**:
 - implementira **vseh 5 modulov naenkrat**,
 - ali najprej uvede **1 modul**.
- Strošek modula: **10 mio €**.
- En modul **ne prinese koristi**, služi kot **pilotni projekt**.
- Sprejetje enega modula odraža **sprejetje celotnega sistema**.

Kaj naredi CIO?



Ste CIO, ki mora izbrati en pristop. **Kako tvegana** se vam zdi uvedba vseh **5 modulov naenkrat**?

Pristop z opcijami pri naložbah v IT (3/3)



NPV (1/2)

- Podjetje razmišlja o uvedbi novega **sistema ERP**, ki bi nadomestil njihov trenutni zastareli sistem.
- Ocenjeni strošek novega ERP sistema je **500.000 €**, pričakuje pa se, da bo sistem v naslednjih **petih letih** prinesel **100.000 € letnih prihrankov**.
- Diskontna stopnja podjetja znaša **10 %**.

$$PV = \frac{R_t}{(1+i)^t}$$

$$NPV(i, N) = \sum_{t=0}^N \frac{R_t}{(1+i)^t}$$

NPV (2/2)

Leto	Sedanja vrednost (PV)
1.	$\frac{100.000 \text{ €}}{(1+0,1)^1} = 90.909 \text{ €}$
2.	$\frac{100.000 \text{ €}}{(1+0,1)^2} = 82.644 \text{ €}$
3.	$\frac{100.000 \text{ €}}{(1+0,1)^3} = 75.131 \text{ €}$
4.	$\frac{100.000 \text{ €}}{(1+0,1)^4} = 68.301 \text{ €}$
5.	$\frac{100.000 \text{ €}}{(1+0,1)^5} = 62.069 \text{ €}$

$$\begin{aligned}\mathbf{NPV} &= 90.909 + 82.644 + 75.131 + 68.301 \\ &+ 62.069 - 500.000 = -120.946 \text{ €}\end{aligned}$$

- **Metoda točk primerov uporabe** (*angl. Use Case Points (UCP)*)
- postopek:
 1. Opredelitev primerov uporabe in **diagramov primerov uporabe** (*angl. Use Case Diagram*),
 2. Izračun **neprilagojene uteži akterjev** (*angl. Unadjusted Actor Weight (UAW)*),
 3. Izračun **neprilagojene uteži primerov uporabe** (*angl. Unadjusted Use-Case Weight (UUCW)*),

- postopek:

- ④ Ocena **faktorja tehnične kompleksnosti** (*angl. Technical Complexity Factor (TCF)*) in **faktorja okoljske kompleksnosti** (*angl. Environmental Complexity Factor (ECF)*).
- ⑤ Izračun **točk primerov uporabe** (*angl. Use-Case Points (UCP)*)

$$UCP = (UAW + UUCW) \cdot TCF \cdot ECF$$

- postopek:

- ⑥ Izračun **napora** (*angl. effort*)

$$\text{napor} = UCP \cdot PF \text{ (faktor produktivnosti)}$$

- ⑦ Izračun **stroška** (*angl. cost*)

$$\text{strošek} = \text{napor} \cdot \frac{\text{strošek}}{h}$$

Pridobivanje IS

Strategije pridobivanja IT aplikacij

- **Koliko kode** razviti?
 - Obstoječa aplikacija, prilagoditev aplikacije, lastni razvoj.
- **Način financiranja?**
 - Nakup, najem.
- **Kje** bo aplikacija **delovala?**
 - Lastna infrastruktura, oblak/SaaS.
- **Od kod** aplikacija **prihaja?**
 - Dobavitelj, odprtokodna rešitev, zunanji razvoj (*angl. outsourcing*).



Kaj je običajno najprimernejša izbira za **startup z malo kapitala** in potrebo po **hitri uvedbi** rešitve?

Nakup obstoječe aplikacije (*angl. Buy-of-the-shelf*)

Prednosti

- **Hitrejša uvedba,**
- **nižji razvojni stroški,**
- **možno preizkusiti pred nakupom,**
- **znane funkcionalnosti in zmogljivosti.**

Slabosti

- **Nepopolno ujemanje s potrebami,**
- **težavna prilagoditev,**
- **odvisnost od dobavitelja,**
- **izzivi pri integraciji z obstoječimi sistemi.**

Prilagoditev ali najem aplikacije

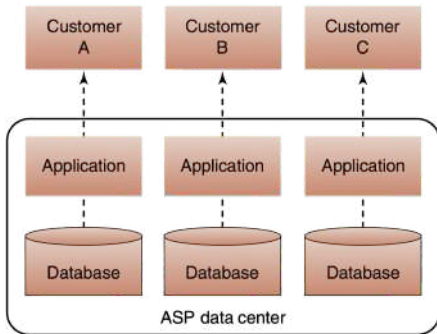
Prilagoditev

- Prilagoditev **programske opreme** **ponudnika**,
- večja **fleksibilnost**,
- lahko **kompleksna in draga**,
- težave pri **nadgradnjah**.

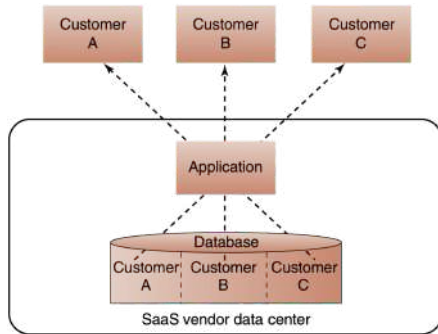
Najem

- **Nižji** **začetni stroški**,
- hitrejša **uvedba**,
- pravilo **80/20**,
- pogosto **SaaS/ASP**.

ASP in SaaS



ASP (*angl. Application Service Provider*)



SaaS (*angl. Software-as-a-Service*)

Odporna koda in zunanje izvajanje

Odpertokodna programska oprema

- **Odpertokodna licenca,**
- možnost **uporabe, prilagoditve** ali **razvoja,**
- alternativa **komercialni programski opremi.**

Zunanje izvajanje (*angl. outsourcing*)

- IT storitve pri **zunanjem izvajalcu,**
- dostop do **strokovnjakov,**
- **nižji stroški** in **hitrejši razvoj,**
- tveganje: **nadzor nad podatki.**

Programske rešitve iz zunanjih virov

TOTAL COST OF OFFSHORE OUTSOURCING

Cost of outsourcing contract

\$10,000,000

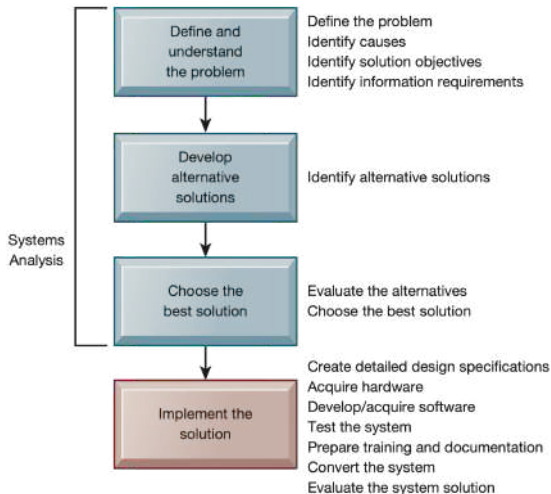
Hidden Costs	Best Case	Additional Cost (\$)	Worst Case	Additional Cost (\$)
1. Vendor selection	0.2%	20,000	2%	200,000
2. Transition costs	2%	200,000	3%	300,000
3. Layoffs & retention	3%	300,000	5%	500,000
4. Lost productivity/cultural issues	3%	300,000	27%	2,700,000
5. Improving development processes	1%	100,000	10%	1,000,000
6. Managing the contract	6%	600,000	10%	1,000,000
Total additional costs		1,520,000		5,700,000
	Outstanding Contract (\$)	Additional Cost (\$)	Total Cost (\$)	Additional Cost
Total cost of outsourcing (TCO) best case	10,000,000	1,520,000	11,520,000	15.2%
Total cost of outsourcing (TCO) worst case	10,000,000	5,700,000	15,700,000	57.0%

Povzetek načinov razvoja informacijskih sistemov

- Življenjski cikel razvoja IS (*angl. Systems Development Life Cycle (SDLC)*),
- prototipiranje,
- razvoj s strani končnih uporabnikov (*angl. **End-user development***),
- programski paketi oz. programska oprema kot storitev (*npr. **SaaS***),
- zunanje izvajanje (*angl. **outsourcing***).

Razvoj IS

Razvoj rešitve informacijskega sistema



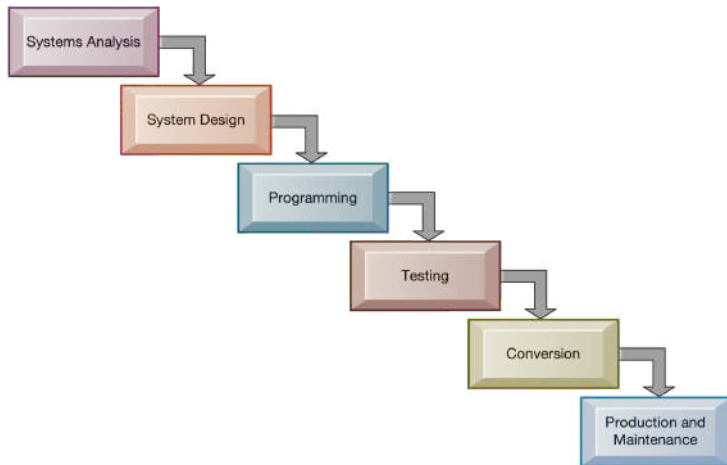
Specifikacija informacijskega sistema

Output	Online reports Hard-copy reports Online queries Order transactions
Input	Online order entry Order status request screen
Database	Database with order file, customer file
Processing	Calculate order totals by type of product. Transmit orders to distribution centers. Track orders by customer. Track orders by sales rep. Schedule deliveries. Update customer data for account changes.
Manual procedures	Sales reps contact customers by phone, email, text message.
Security and controls	Online passwords Only authorized sales reps and company employees can access the system. Only company-owned mobile devices can be used for entering orders or accessing corporate data.
Conversion	Input Input customer data Test system
Training and documentation	System guide for users Online training sessions and tutorials

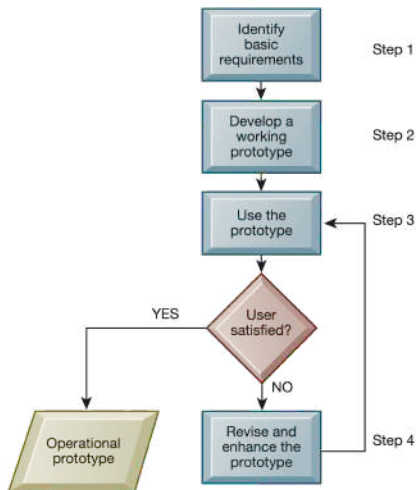
Testiranje informacijskega sistema

Test Case Number: GS02-010	
Prepared by: A. Patel	Date: February 19, 2023
Objective: This subtest checks for an authorized user accessing the system.	
Platform: iOS	
Procedure Description: Select Sign In Select Username Enter Username Select Password Enter Password Select Submit	
Expected Result: When user selects Sign In, the Sign In menu appears When user selects User Name, the cursor moves to the User Name field When user enters that person's system user name the user name appears on the screen When user selects Password, the cursor moves to the Password field. When user enters that person's password, the password appears on the screen as asterisks When user selects Submit, the system verifies the entered data and allows the user to access the system When user enters an incorrect (or unauthorized) username or password, the error message "Wrong User Name or Password" appears	
Test Results: All OK	

Tradicionalni razvojni cikel (SDLC)



Prototipiranje



Primer: avtomatizirano testiranje programske opreme pri McAfee

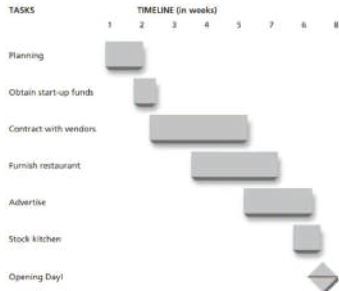
- Podjetje: McAfee (varnostna programska oprema),
- Okolje: SAP ERP, več kot **40 poslovnih sistemov**,
- Razvoj: prehod na **agilni razvoj (2-tedenski sprinti)**,
- Problem: preveč sprememb → **ročno testiranje prepočasno**,
- Rešitev: **avtomatizirano testiranje (Worksoft Certify)**
- Rezultat: ~ **2.500 ur** in **\$200.000 prihranka** ter bolj zanesljivo testiranje kompleksnih procesov

Projektno vodenje IS

Kaj je projekt?

A project

- ▶ is a temporary endeavor
- ▶ has a specific, unique purpose
- ▶ has a primary customer or sponsor, as well as stakeholders
- ▶ requires resources, staff time, and expertise from different areas
- ▶ includes an element of uncertainty
- ▶ has metrics for success



PROJECT: Opening a new restaurant



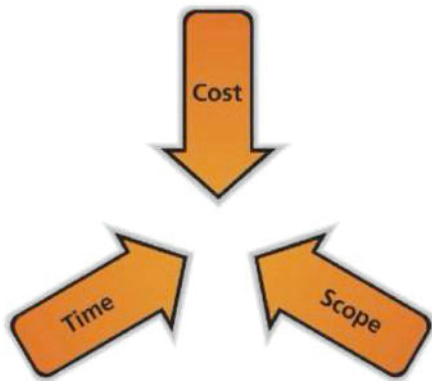
PROCESS: Procuring office supplies

Projekt ...



Kaj od naštetega je **projekt** in ne proces?

Trojna omejitev projekta

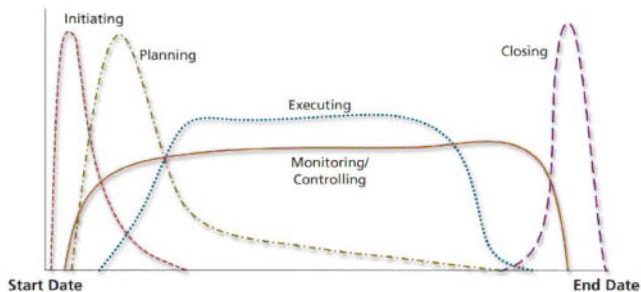


- Začasen,
- unikaten cilj,
- omejeni viri,
- negotovost,
- merila uspeha.

Procesi projektnega vodenja

• 5 procesov:

- Inicijacija (*angl. Initiating*), načrtovanje (*angl. Planning*), izvajanje (*angl. Executing*), spremljanje in nadzor (*angl. Monitoring & Controlling*), zaključek (*angl. Closing*).



Načrtovanje projekta

Project: Social Media Campaign for The Wild River Waterslide Ride

Start and End Dates: April 2–June 30

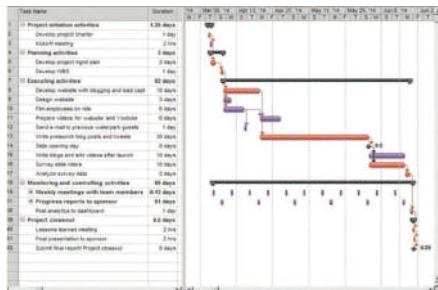
Key Participants:

- Project Manager: Stan M.
- Sponsor: Consuela T., Director of Marketing
- Estimated budget: \$38,000 (includes personnel salaries)
- Team Members:
 - Bailey L., Programmer/Analyst
 - John C., Copywriter
 - Lucinda V., IT Support

Project Objectives: Develop, launch, and track a social media campaign to promote the waterslide throughout the region. The goal is a 20% increase in number of park visitors from June 15 to July 15, compared to the same period last year. Additional metrics will include (1) web analytics on the blog and other online assets, such as percent change in the number of video views and unique visitors, and (2) survey of Wild River riders to determine how they learned about the ride.

Project Approach: With IT's support, launch a blog containing comments, images, and videos of employees and guests who experience the ride before opening. Post videos to YouTube and distribute links to popular bookmarking sites. Invite loyal customers who maintain their own blogs to a "Wild River Preview" day. Capture leads from the blog by offering discount coupons.

Signatures:

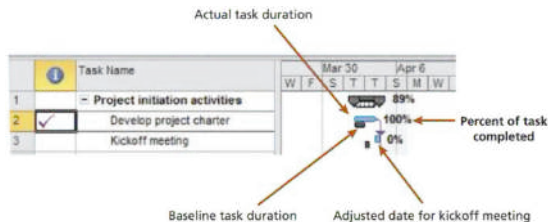


Projektni vodja

Area	Description	Sample Outputs
Integration	Overall management of the project	Project charter; project management plan
Scope	Clarifying what work is part of the project, and what is not	Scope statement; work breakdown structure; scope management plan
Time	Ensuring the project meets its deadlines	Project schedule; Gantt chart; critical path analysis; time management plan
Cost	Determining the budget, analyzing any need for changes, and ensuring the project's costs are tracked and managed properly	Cost estimate; running expenses; net present value analysis; return on investment analysis; cost management plan
Quality	Ensuring the project deliverables meet quality standards for the project	Quality standards; project metrics; quality management plan
Human resources	Managing the people involved in the project, motivating stakeholders, allocating tasks, monitoring workloads	Project organizational chart; assignment matrix; human resources management plan
Communications	Providing updates and status reports to stakeholders; presenting project to sponsor(s)	Status reports; meetings; project dashboard; website; communications management plan
Risk	Managing risks that may impact the project; implementing mitigation strategies	Risk management plan; risk register

- Strong leadership skills
- Excellent communication abilities
- Outstanding "people" skills
- Technical competence in project area
- Good listening skills
- Strong team-building skills
- Excellent presentation skills
- Good problem-solving and critical-thinking skills
- Ability to balance priorities, stay organized, and keep the team on track

Programska oprema za projektno vodenje



Wild River Waterslide Project: Budget Report

Task	Baseline Cost	Actual Cost	Variance
Develop project charter	1,280.00	1,920.00	640.00
Kickoff meeting	670.00	670.00	0
Develop project management plan	1,280.00	1,280.00	0
Develop WBS	640.00	640.00	0
Develop website with blogging capability	8,800.00	11,440.00	2,640.00

Ganttov diagram

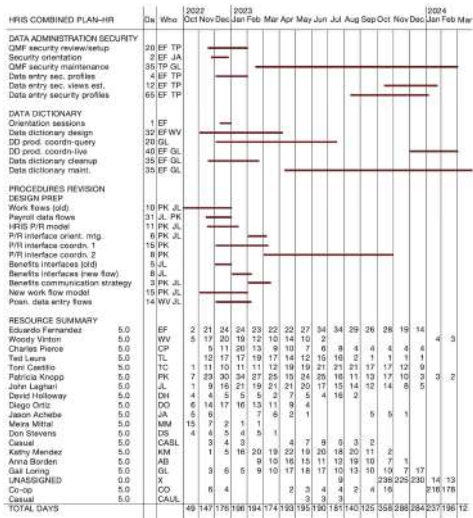
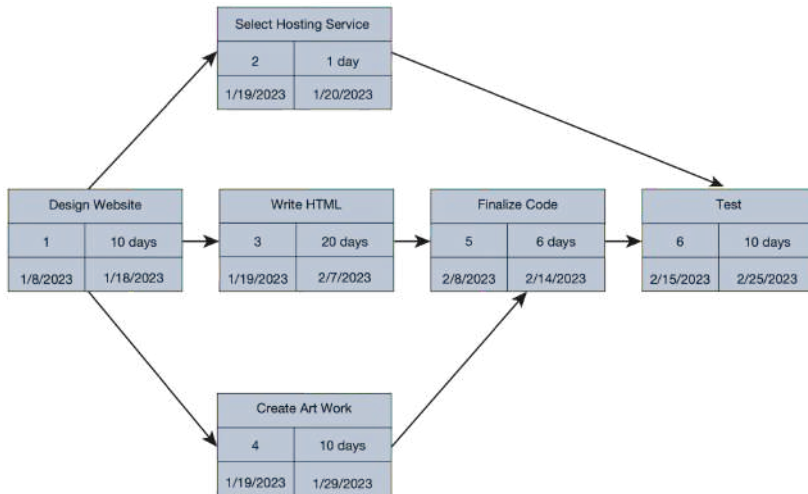


Diagram PERT



Implementacija ERP sistema: **AI-Asasyah**



- Zamenjava podedovanih (*angl. legacy*) sistemov,
- ERP implementacija,
- fazni pristop,
- prototipiranje,
- migracija podatkov,
- uspešna implementacija.

Uspešnost projektov informacijskih sistemov

- Visoka **stopnja neuspešnih projektov IS**,
- napačna ocena **poslovne vrednosti**,
- podcenjevanje **organizacijskih sprememb**,
- potreba po **projektne vodenju**,
- skrbna **izvedba projekta**.

Zakaj projekti uspejo ali propadejo?

Does the project have ...	
People factors	Executive support and sufficient resources allocated? Talented and motivated personnel assigned to the project? Leaders with project management expertise? A strategy to manage conflicts among stakeholders?
Organizational factors	Involvement and buy-in from a broad range of end users? An objective that is perceived to be aligned with business goals?
Project factors	A clear objective and a well-defined scope? A design that will allow any new systems to interact with existing systems?
Project management factors	A well-defined system development methodology that is appropriate for the project? Appropriate project management software and other tools? A clear change control process for managing scope creep? A strategy for assessing quality? A clear process for monitoring and tracking progress?
External factors	Well-understood and binding agreements with vendors and consultants?

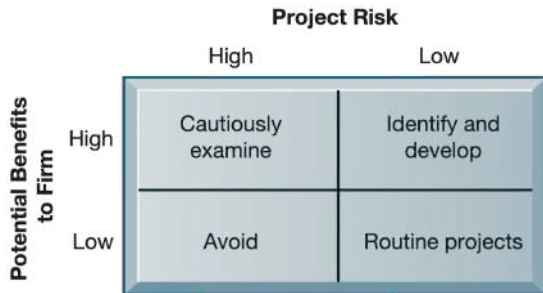
Portfelj, tveganja in strateška negotovost

- **Portfelj projektov** = vsi IT projekti v organizaciji,
- cilj: izbrati **prave projekte**,
- stalni proces:
 - izbor projektov,
 - spremljanje napredka,
 - ustavitev neuspešnih projektov.

Kriteriji za izbiro projektov

Criterion	Example
Contribution to competitive strategy	Projects that advance the organization's competitive position should be rated highly. One example is Carfax.com's web-based software to access a used car's history of accidents or defects based on the vehicle identification number.
Return on investment	Projects that offer a substantial return on investment earn high ratings. For example, Oregon Corrections switched to Voice over IP for its call center, staffed by prison inmates, reducing monthly telecom charges by 40%.
Compliance and risk reduction	Reducing risk and ensuring proper compliance are especially important drivers in some industries. Companies in the financial services industry spend more than 8% of their net income on IT projects and services, much of it to ensure compliance and mitigate risk.

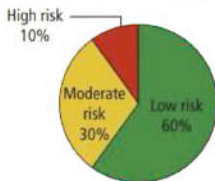
Analiza portfelja projektov IS



Spregljanje portfelja projektov

Projects	% Completed	Priority	Expense to Date	Risk Rating
Mobile Device Encryption	15%	1	\$157,844	1
M-commerce initiative	69%	2	\$258,333	5
Mobile App	20%	1	\$15,006	5
Mobile Device Management	88%	2	\$58,774	2
Social Media Presence	100%	1	\$5,688	3
Data Analytics Project	0%	3	\$0	5

Portfolio Status, May 25



Politike uporabe informacijskih sistemov

- ▶ What IT services are provided to each employee?
- ▶ Can employees access the Internet?
- ▶ Is access to certain websites prohibited?
- ▶ What kind of surveillance and monitoring is in place?
- ▶ Are employees permitted to use the company's IT resources for personal use? If so, what are the constraints?
- ▶ How does the organization respond to violations? What are the penalties?
- ▶ Can employees download and install software without IT approval?
- ▶ Can employees send mass email to colleagues about personal issues, such as "kittens are ready to adopt" or "donuts available"?
- ▶ Can employees use IT resources for fund-raising or commercial ventures?
- ▶ Can employees use the company name when they post on blogs or other social media?
- ▶ Security requirements
 - ▶ Password construction and change schedules
 - ▶ Encryption requirements
 - ▶ Prohibition against sharing accounts
 - ▶ Confidentiality requirements

Pravna tveganja in e-poštna opozorila

Potential Threat	Purpose of Disclaimer
Breach of confidentiality	The disclaimer can warn recipients against disclosing information contained in the email.
Accidental breach of confidentiality	If the email contains confidential information intended only for the recipient, a third party (such as the postmaster) might receive it accidentally. Wording that the message is intended only for the addressee provides some protection.
Transmission of viruses	The disclaimer can warn recipients that they are responsible for checking the message for viruses.
Implied contractual obligations	An employee's email might contain wording that implies a firm contract. The disclaimer can add that any contract must be confirmed by the employee's manager.
Negligent misstatements	The company may be liable if an employee provides advice that the recipient then relies on. The disclaimer can include wording to protect the company from this.
Employee misuse	Absent a disclaimer, the employer may be held responsible for an employee who misuses email and violates the company's policies, for example by making libelous or defamatory statements in email.

Obnova po nesreči in neprekinjeno poslovanje

- **Obnova po nesreči** (*angl. Disaster recovery*),
 - postopki za obnovo IT infrastrukture po nesreči,
- **neprekinjeno poslovanje** (*angl. Business continuity*),
 - zagotavljanje neprekinjenega poslovanja,
- ključni elementi:
 - varnostne kopije podatkov,
 - nadomestna lokacija (*angl. recovery site*),
 - postopki ponovnega zagona sistemov,
 - testiranje načrtov.

Vrste nesreč in posledice

Types of Disasters	Possible Consequences
Natural disasters (floods, fires, hurricanes, snowstorms, earthquakes, volcano eruptions)	Electrical outages, destruction of technology infrastructure; disruption of supply chains
Security breaches (unauthorized break-ins, employee theft, denial of service attacks, information leaks)	Release of confidential information, disabled servers, corruption or alteration of data, malware infections
Pandemics	Massive absenteeism
Environmental hazards (hazardous materials spills, radiation, letters containing white powder)	Evacuation; system disruptions; inability of staff to work in the company's facilities

Tehnološki trendi

Trend	Description
Autonomous devices	Devices that use artificial intelligence to perform tasks traditionally done by humans, including robots, autonomous vehicles, drones, and intelligent agents
Augmented analytics	The use of automated algorithms to explore hidden patterns and relationships in large amounts of data
Blockchain	A distributed electronic ledger that is shared by all participants in the network and difficult to hack
Digital ethics and privacy	The emphasis by consumers on the growing importance and value of their personal information

Prihodnji tehnološki trend



Kateri tehnološki trend bo po vašem mnenju v **naslednjih 5 letih najbolj vplival** na organizacije?

Človeški dejavnik pri strateškem načrtovanju

- Strateško načrtovanje temelji na **človeških odločitvah**,
- analitiki ocenjujejo: trende, tehnologije in trg,
- napovedovanje prihodnosti je **negotovo**,
- na odločitve pogosto vplivajo: izkušnje, intuicija, pristranskosti.

Napačne tehnološke napovedi

"There is no reason anyone would want a computer in their home."—Ken Olson, president, chairman, and founder of Digital Equipment, mini and mainframe computer manufacturer, 1977.

"I have traveled the length and breadth of this country and talked with the best people, and I can assure you that data processing is a fad that won't last out the year."—The editor in charge of business books for Prentice Hall, 1957.

"I think there is a world market for maybe five computers."—Thomas Watson, chairman of IBM, 1943.

"The wireless music box has no imaginable commercial value. Who would pay for a message sent to no one in particular?"—Associates of David Sarnoff responding to the latter's call for investment in radio in 1921.

"The Americans have need of the telephone, but we do not. We have plenty of messenger boys."—Sir William Preece, chief engineer, British Post Office, 1878.

"This 'telephone' has too many shortcomings to be seriously considered as a means of communication. The device is inherently of no value to us."—A memo at Western Union telegraph company, 1878.

"When the Paris Exhibition closes, electric light will close with it and no more will be heard of it."—Erasmus Wilson, Oxford professor, 1878.

Kognitivne pristranskosti

Cognitive Bias	Description
Confirmation bias	The tendency to choose information to review that supports our existing position and ignore conflicting evidence
Overconfidence	The act of having more faith in our own estimates than is realistically justified
Planning fallacy	The tendency to underestimate the time it will take to complete a task
Anchoring	Reliance on one piece of information, however irrelevant
Availability bias	The tendency to judge the probability of an event based on how easily examples come to mind
Hindsight bias	The belief that an actual event was predictable even if it was not

Dogodki črnega laboda



(a)



(b)



Zaključek

Kratek kviz

